Image Deblurring and Object Detection using Wiener Filter and YOLOv8n Model

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Abstract

This project investigates the effect of image blur on object detection and evaluates whether classical deblurring methods improve downstream performance. Gaussian filtering and Wiener deconvolution were implemented and assessed using PSNR and SSIM. A pretrained YOLOv8n detector was first used to compare blurred, restored, and original images. A corrected YOLO training pipeline was then built using a 300-image subset of COCO and COCOBlur data. Results show that Wiener filtering slightly improved image-quality metrics, but the restored images did not improve downstream detection over blurred inputs. In the final trained-model evaluation, original sharp images achieved the strongest performance, followed by blurred images, while deblurred images performed worst.

1. Introduction

Image blur caused by motion or defocus reduces visual quality and negatively affects object detection. This project studies whether classical deblurring can improve image quality and whether that improvement helps object detection. The project combines restoration, pretrained detection analysis, dataset preparation, model training, and critical evaluation.

2. Step 1: Version Control and AI-Assisted Coding

Git was used to manage project files and maintain version control. A GitHub repository was created for the final code and documentation. Generative AI tools were used to assist with workflow planning, debugging, and drafting documentation. All generated code and text were manually reviewed, modified, and tested before inclusion.

3. Step 2: Image Deblurring

3.1 Methods

Two classical restoration methods were evaluated:

- Gaussian filtering as a baseline

- Wiener deconvolution as a restoration-based method

3.2 Metrics

Image quality was measured using:

- PSNR

- SSIM

3.3 Results

Method	PSNR	SSIM
Gaussian	23.009	0.650
Wiener	23.147	0.663

3.4 Discussion

Wiener filtering slightly outperformed Gaussian filtering on PSNR and SSIM. However, the improvement was modest, showing that better low-level restoration quality did not necessarily imply stronger downstream performance.

4. Step 3: Object Detection and Analysis

4.1 Detector

A pretrained YOLOv8n model was used.

4.2 Compared inputs

- Blurred images
- Gaussian-filtered images
- Wiener-restored images

4.3 Results

Input	Avg. Detections	Avg. Confidence
Blurred	5.0	0.471
Gaussian	5.0	0.494
Wiener	4.0	0.413

4.4 Discussion

Gaussian filtering slightly improved average confidence, while Wiener restoration reduced average detections and confidence. This shows that improved restoration metrics alone did not improve pretrained detector behavior.

5. Step 4: Training Data Preparation and Model Training

5.1 Dataset preparation

A subset of 300 images was created from COCO 2017 validation data and COCOBlur:

- Train: 180
- Validation: 60
- Test: 60

COCO annotations were converted into YOLO label format.

5.2 Training issue and fix

An important issue was discovered during training: the detector initially produced zero box loss and zero mAP because the dataset structure did not follow the standard YOLO `images/labels` layout. After reorganizing the data into:

- `images/train`, `images/val`, `images/test`
- `labels/train`, `labels/val`, `labels/test`

the model began training correctly.

5.3 Training setup

- Model: YOLOv8n
- Epochs: 100
- Image size: 640
- Hardware: NVIDIA RTX 4050 Laptop GPU

5.4 Training observations

After the dataset fix, the model trained successfully. Early validation results showed meaningful performance:

- Epoch 1 mAP50: 0.613
- Epoch 1 mAP50-95: 0.453

Later epochs fluctuated around lower but still meaningful values, confirming that the corrected pipeline was functioning.

6. Step 5: Performance Comparison and Critical Analysis

6.1 Final trained-model results

| Input | Avg. Detections | Avg. Confidence |

|---|---:|---:|

| Blurred | 30.983 | 0.108 |

| Deblurred | 23.217 | 0.113 |

| Original | 58.15 | 0.161 |

6.2 Analysis

The final trained model performed best on original images, confirming that sharp images remain the most informative input for detection. Blurred images outperformed the deblurred images in detection count, while the deblurred images showed only a very small confidence increase over blurred images. This suggests that the chosen classical deblurring method improved low-level image metrics but did not improve the detector's downstream performance.

6.3 Key finding

A major conclusion of this project is that better image restoration quality does not automatically lead to better object detection.

6.4 Limitations

- Small training subset (300 images)

- Classical deblurring methods only

- No deep learning restoration models

- Final evaluation based on average detections and confidence rather than full COCO test metrics

6.5 Future work

- Evaluate deep learning-based restoration methods

- Use larger datasets

- Compare more detectors

- Explore joint restoration-and-detection training

7. Conclusion

This project demonstrates that classical deblurring methods can slightly improve image-quality metrics, but those improvements do not guarantee better object detection performance. Wiener filtering slightly improved PSNR and SSIM, but did not improve detector performance compared with blurred inputs. After correcting the YOLO dataset structure, training became successful and the final model showed the strongest performance on original sharp images, followed by blurred images, with deblurred images performing worst. The study highlights the importance of evaluating restoration methods using downstream tasks rather than image metrics alone.

8. AI Usage Statement

Generative AI tools were used for workflow planning, debugging, and documentation drafting. All generated suggestions and code were manually reviewed, tested, and adapted before inclusion.

9. References

- COCO 2017 Dataset

- COCOBlur Dataset

- Ultralytics YOLOv8 Documentation